

AI/ML intern DAY 11

TASK — AI Model Exploration

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Introduction

The objective of this task was to study Artificial Intelligence models used in Virtual Try-On Systems, Human Pose Estimation, Body Segmentation, and Fashion Recommendation technologies.

The research focused on:

- human pose estimation
- body landmark detection
- segmentation models
- AI fashion recommendation integration

The task helped understand how modern AI systems analyse human body structure and generate intelligent virtual fitting workflows.

Models Studied

Model	Purpose
MediaPipe Pose	Human pose estimation
Detectron2	Human segmentation
U2Net	Background removal
OpenPose	Human body tracking

1. MediaPipe Pose

Purpose

MediaPipe Pose is an AI-based human pose estimation framework developed for:

- body landmark detection
- skeletal tracking
- posture analysis
- body alignment

Features Studied

- shoulder tracking
- elbow tracking
- hip tracking
- knee detection
- full body landmark analysis

Applications

- virtual try-on systems
- fitness tracking
- gesture analysis
- posture correction
- motion tracking

Basic MediaPipe Pose Detection Code

Save file as:

pose_detection.py

Python Code

```
import cv2
import mediapipe as mp

img = cv2.imread("person.jpg")

mp_pose = mp.solutions.pose
mp_draw = mp.solutions.drawing_utils

pose = mp_pose.Pose()

rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

results = pose.process(rgb)

if results.pose_landmarks:
```

```

mp_draw.draw_landmarks(
    img,
    results.pose_landmarks,
    mp_pose.POSE_CONNECTIONS
)

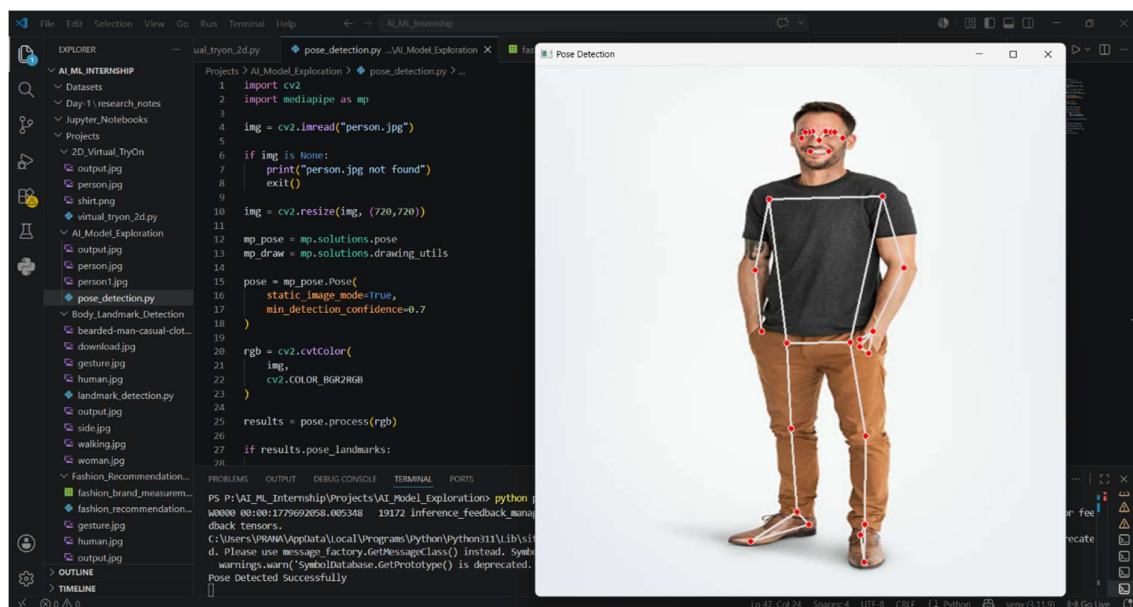
print("Pose Detected")

cv2.imshow("Pose Detection", img)

cv2.waitKey(0)

cv2.destroyAllWindows()

```



2. Detectron2

Purpose

Detectron2 is an advanced AI segmentation framework developed for:

- human segmentation
- object detection
- body masking
- region extraction

Features Studied

- human parsing

- clothing segmentation
 - object detection
 - body region extraction
-

Applications

- virtual try-on systems
 - AI fashion analysis
 - image segmentation
 - body masking workflows
-

3. U2Net

Purpose

U2Net is an AI model used for:

- background removal
 - human segmentation
 - image masking
 - object extraction
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Features Studied

- background removal
 - cloth extraction
 - foreground segmentation
 - image masking
-

Applications

- AI fashion editing
- virtual fitting systems
- image preprocessing
- cloth extraction workflows

4. OpenPose

Purpose

OpenPose is an AI framework used for:

- multi-body pose estimation
- skeletal tracking
- gesture recognition
- motion analysis

Features Studied

- body keypoint detection
- human motion tracking
- multi-person detection
- skeletal visualization

Applications

- motion capture systems
- virtual avatars
- fitness analysis
- AI movement tracking

Human Pose Estimation Study

Topics Covered

- body landmark detection
- skeletal tracking
- posture analysis
- body alignment systems
- gesture recognition

The research focused on how AI identifies human body joints and body movement patterns.

Body Landmark Detection

Landmarks Analysed

- shoulders
- elbows
- wrists
- hips
- knees
- ankles

The AI models use landmark coordinates for:

- body tracking
 - motion analysis
 - virtual garment fitting
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AI Fashion Recommendation Integration

Studied how AI models integrate with:

- outfit recommendation systems
 - customer preference analysis
 - smart fashion systems
 - virtual fitting platforms
-

Segmentation Model Research

Segmentation Areas Studied

- human parsing
- body masking
- cloth segmentation
- foreground extraction
- garment isolation

The research improved understanding of how AI separates body and clothing regions in fashion applications.

AI Workflow Architecture

Input Human Image



Pose Detection



Body Landmark Extraction



Segmentation Processing



Garment Alignment



AI Recommendation



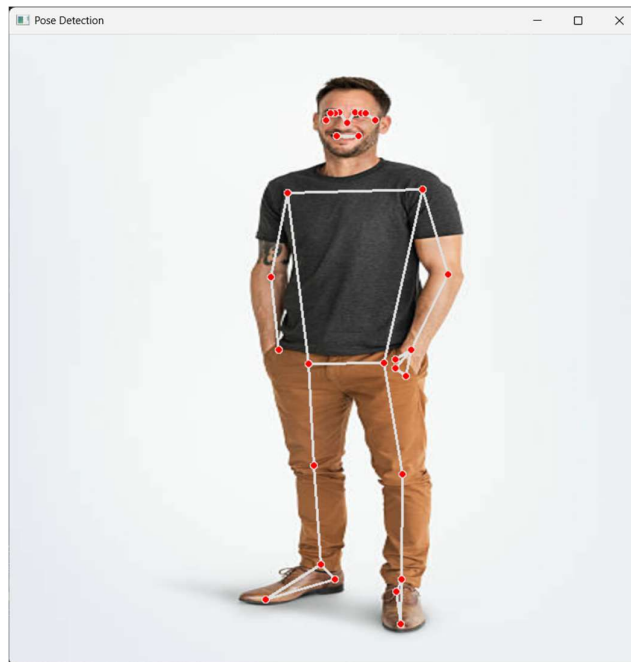
Virtual Try-On Output

Technologies Used

- Python
 - OpenCV
 - MediaPipe
 - Detectron2
 - U2Net
 - OpenPose
 - Artificial Intelligence
 - Machine Learning
 - Computer Vision
 - Deep Learning
-

Expected Outputs

- pose estimation
- body landmark visualization
- skeletal tracking
- segmentation analysis
- body masking workflows
- AI recommendation research



Research Findings

The research helped understand:

- AI body tracking systems
- human segmentation workflows
- skeletal landmark analysis
- body masking technologies
- AI recommendation integration

The task demonstrated how AI models improve virtual fitting systems using Computer Vision and Deep Learning workflows.

Learning Outcomes

During this task, the following concepts and skills were learned:

- Human pose estimation
- skeletal landmark tracking
- body segmentation workflows
- AI background removal
- motion tracking systems
- AI recommendation integration

- human parsing technologies
 - computer vision applications
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Conclusion

The AI Model Exploration task successfully analysed modern Artificial Intelligence models used in Virtual Try-On Systems, Human Segmentation, and Fashion Technology applications.

The task improved understanding of:

- pose estimation systems
- body landmark detection
- segmentation workflows
- AI recommendation integration
- skeletal tracking systems
- virtual fitting technologies
- computer vision applications